① a) $\boxed{T = 324°F}$

$500 = 298.15 + x(807.29)$

$\boxed{x = 0.2499}$

$v = 0.01774 + x(4.4327 - 0.01774)$

$\boxed{= 1.12\ ft^3/lbm}$

$h = 298.51 + 0.2499(888.99)$

$\boxed{= 521\ Btu/lbm}$

b) $\boxed{p = 73.4\ lbf/in^2}$

$v = 0.032 + 0.5(3.9680 - 0.02503)$

$\boxed{= 2\ ft^3/lbm}$

$u = 486.5 + 0.5(495 - 486.5)$

$\boxed{= 490.75\ \cancel{Btu}/lb}$

c)

$T = 450 + \dfrac{(500-450)(0.04034 - 0.03679)}{(0.04175 - 0.03817)}$ $\boxed{= 440.30°C}$

$u = 2966.7 + \dfrac{(3064.3 - 2966.7)(0.04034 - 0.03817)}{(0.04175 - 0.03817)}$ $\boxed{= 3025.86\ kJ/kg}$

$h = 3272 + \dfrac{(3348.3 - 3272)(0.04034 - 0.03817)}{(0.04175 - 0.03817)}$ $\boxed{= 3349.53\ kJ/kg}$

D) $p = 1.3268\ bar$

$235.31 = 173.82 + x(386.66 - 173.82)$

$\boxed{x = 0.289}$

$v = 0.0007384 + 0.244(0.1474 - 0.000734)$

$\boxed{= 0.0043\ m^3/kg}$

$u = 235.31 - (1.3264 \times 10^5 \times 0.043)$

$\boxed{= 229.54\ kJ/kg}$

②

A) $\boxed{P_1 = 1.0022\ MPa}$

$V_1 = V_2 = 0.14405\ m^3/kg$

$V_{f2} = 1.0435 \times 10^{-3}\ m^3/kg$

$V_{g2} = 1.673\ m^3/kg$

b) $x_2 = \dfrac{0.14405 - 1.0435 \times 10^{-3}}{1.673 - 1.0435 \times 10^{-3}}$

$\boxed{x = 0.1154}$

③

a) $\boxed{T_1 = 133.6°C}$

b) $v_2 = \frac{2.034\,m^3}{2\,kg} = 1.017\,m^3/kg$ $\qquad T_2 =$

④

$$\frac{T_1 - T_2}{T_1 - T_x} = \frac{v_1 - v_2}{v_1 - v_x}$$

⑤ a) $\boxed{31.2\,kPa}$ b) $Q = 1.5(2530 - 2469.6)$

$\boxed{Q = 90\,kJ}$

60%

⑥

a) $v = 0.4352\times10^{-3} + 0.6(0.0236 - 0.4352\times10^{-3})$

$= 0.01449\,m^3/kg$ $\qquad 0.6 = \frac{m_g}{m_g + 25}$ $\qquad m_g = 37.5\,kg$

$V_g = 0.0236 \times 37.5 = 0.885\,m^3$ $\qquad V = (25 + 37.5)\,0.01449 = \boxed{0.9056\,m^3}$

$V_f = 25 \times 0.4352\times10^{-3} = 0.01088\,m^3$

b)

$\%vol = \frac{0.885}{0.9056} \times 100 = \boxed{97.725\%}$

(7) $P_1 = 1.3\ bar$ $P_2 = 0.8\ bar$ $T_1 = 27°C = 300K$ $m = 0.2g$
$= 130kPa$ $= 80kPa$ $T_2 = 17°C = 290k$ $= 2 \times 10^{-4} kg$

$R = \frac{8.314}{4.003}$

$R = 2.077 kJ/kgk$

$P_1V_1 = mRT_1$

$V_1 = \frac{2 \times 10^{-4} \cdot 2.077 \cdot 300}{130} = \boxed{958.6 cm^3}$

$V_2 = \frac{2 \times 10^{-4} \cdot 2.077 \cdot 290}{80} = \boxed{1505.4 cm^3}$

(8) $P_1 = 275kPa$ $V_1 = 0.05m^3$ $T = 298.15k$

$R = 0.189$

$m = \frac{275kPa \cdot 0.05m^3}{0.189 \times 10^3 \cdot 298.15} = \boxed{m = 2.44 \times 10^{-4}}$

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